Elementopia 2.0

Interactive Gamified Chemistry Learning Platform

Cebu Institute of Technology – University | College of Computer Studies

Course / Section: BSIT / 2627-sem1-it411-53
Adviser: Engr. Ralph P. Laviste / Dr. Patrick L. Bacalso
Document: Official Codebase & Repository Verification

CIT-U CAPSTONE | REACT 19 & VITE | SPRING BOOT 3 (JAVA 17) | SUPABASE POSTGRESQL 15 | DEPED GRADE 9 ALIGNED

Official GitHub Repository:	https://github.com/igoirshiy/Elementopia-Main
Live Staging Application:	https://elementopia-game.vercel.app
Repository Visibility & Branch:	**Public** • Primary Branch: `main`

1. Project Development Team (Group 53)

Team Member	Primary Role	Core Engineering Responsibilities
Danielle Maxine P. Ocampo	Project Manager & Lead Dev	Architectural coordination, sprint milestones, synthesis workflow integration.
Clyde C. Benolirao	Full Stack & Gameplay Engineer	Resonance puzzle engine, 1v1 matchmaking lobby, interactive workbench controls.
Jian Marc Ceniza	Backend & Database Architect	Spring Boot 3 REST controllers, Supabase schemas, HikariCP database pooling.
Erica Y. Dabalos	Frontend & UI/UX Specialist	Responsive design, Dr. Atom modals, styling systems, accessible mobile touch bounds.
Wayne Kenji B. Dibdib	QA & Telemetry Specialist	Pre-test cloud sync verification, telemetry schema testing, regression test suite.

2. Technology Architecture

Tier / Layer	Technology	Role & Specifications
Presentation (Client)	React 19, Vite 6, TailwindCSS v4	High-performance reactive UI, touch drag-and-drop (@dnd-kit), canvas animation (Konva).
Logic & Validation	Spring Boot 3.x, Java 17 (Maven)	Reaction controller, stoichiometry validation, Hazmat sliding-window engine.
Data Persistence	Supabase (PostgreSQL 15)	Cloud database for session telemetry, diagnostic pre-tests, and compound blueprints.
Realtime Network	Supabase Realtime (WebSockets)	Sub-second peer synchronization, 3-sec heartbeats, and graceful disconnect engine.
Deployment & CI/CD	Vercel (Edge) & Render (Docker)	Automated continuous deployment for SPA client and containerized backend API.

3. Repository Directory Structure

```
Elementopia-Main/
├── .github/                  # Automated GitHub Actions CI/CD workflows
├── Documents/                # Capstone Documentation (SRS v2.0, SDD v2.0, STD, MVP Validation)
│   ├── Refactored_Elementopia_SRS.pdf
│   ├── Refactored_Elementopia_SDD.pdf
│   └── MVP Validation Findings & Refactoring Priorities.pdf
├── Elementopia/
│   ├── BACKEND/              # Spring Boot 3 + Java 17 REST API Service
│   │   ├── src/main/java/    # Controllers, Models, Services, Progression logic
│   │   ├── Dockerfile        # Container deployment configuration for Render
│   │   └── pom.xml           # Maven build configuration and dependencies
│   ├── FRONTEND/             # React 19 + Vite Web Application
│   │   ├── src/features/     # resonance-puzzle, 1v1-challenge, mastery-dashboard
│   │   ├── vercel.json       # Production SPA rewrite routing configuration
│   │   └── package.json      # Node package manager dependencies & scripts
│   ├── start-backend.bat     # Windows one-click backend development launcher
│   └── start-frontend.bat    # Windows one-click frontend development launcher
└── README.md                 # Comprehensive open-source documentation
```

4. Core Modules & Refactored Capabilities (v2.0)

Module 1: Elemental Resonance Puzzle

- **Stoichiometric Synthesis:** Drag-and-drop valence electron pairing to form ionic and covalent bonds.
- **Meaningful Byproduct Feedback:** Real-time diagnostic alerts when an invalid mix yields real chemical compounds.
- **Hazmat Protocol Anti-Guessing:** Sliding-window algorithm that temporarily locks workbench upon rapid spam clicking.
- **Two-Tier Educational Debrief:** Explicit milestone reviews explaining polar/nonpolar covalent vs ionic transfer.

Module 2: Assessment & Progress

- **Automated Telemetry Sync:** Asynchronous dispatch of pre-test baselines directly to Supabase cloud within < 2s.
- **Learning Gain Tracking:** Automated calculation of Hake's normalized learning gain ($g \geq 0.40$).
- **Visual Mastery Dashboard:** Radar charts and progress indicators tracking student mastery of DepEd competencies.

Module 3: 1v1 Opponent Challenge

- **5-Digit Room Matchmaking:** Fast peer connection in < 5 seconds via WebSocket channels.
- **Synchronized Synthesis Race:** Live sub-second race to synthesize target chemical compounds.
- **Heartbeat Disconnect Engine:** 3s ping + 8s silence detector triggers 15s grace modal and clean forfeit win resolution.

Non-Functional Usability Standards

- **Mobile Touch Target Standards:** Enforces ≥ 48px × 48px interactive boundaries across 360px–768px viewports.
- **Zero-Friction Access:** Instant guest nickname onboarding without requiring email registration.

5. Local Development Quick-Start

```
# 1. Clone the verified repository:
git clone https://github.com/igoirshiy/Elementopia-Main.git
cd Elementopia-Main

# 2. Run the Frontend (React 19 + Vite):
cd Elementopia/FRONTEND
npm install
npm run dev                  # Serves on http://localhost:5173

# 3. Run the Backend (Spring Boot 3 + Java 17):
cd ../BACKEND
.\mvnw.cmd clean spring-boot:run   # Serves on http://localhost:8080
```

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